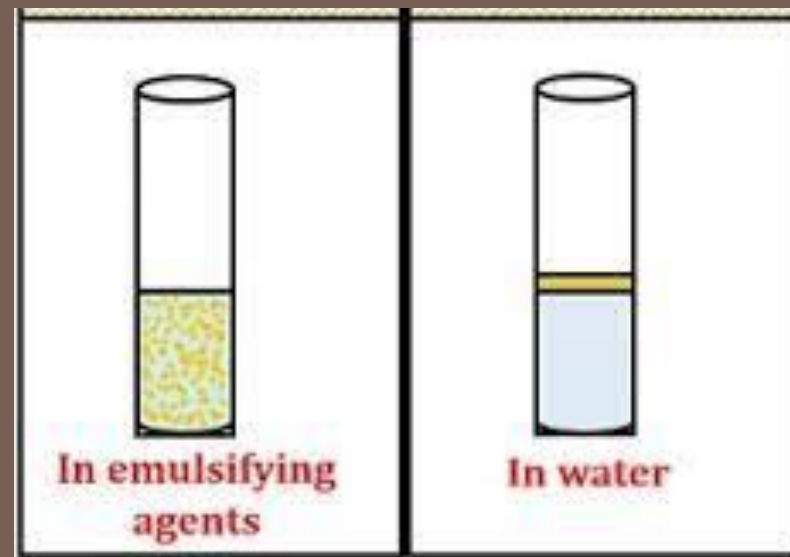


EMULSIFICATION TEST & HAY'S TEST



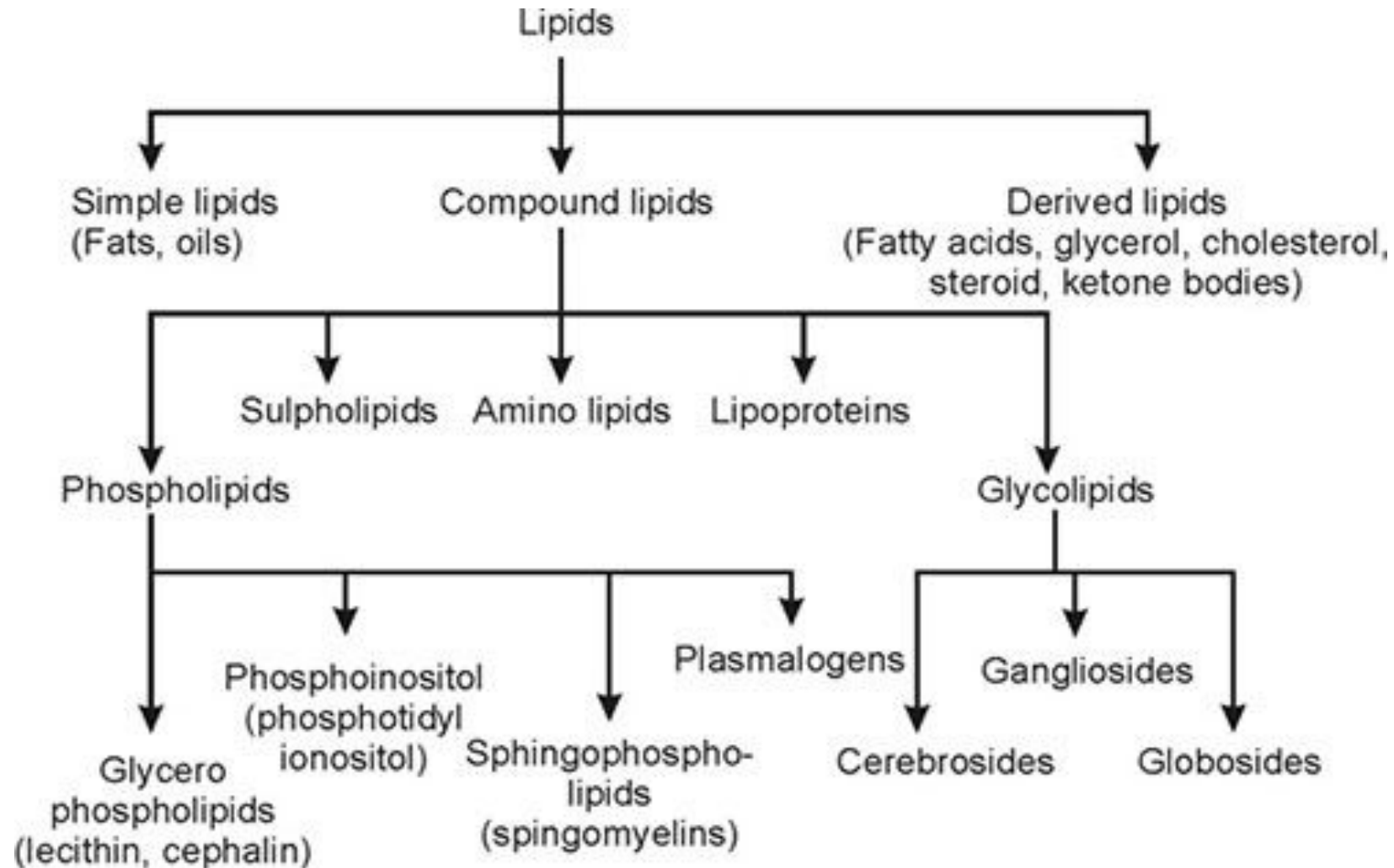
Introduction of Lipids

- The lipids are organic substances occurring in plant and animal tissues and belong to a very **heterogeneous group of substances** which have only a few properties in common.
- Lipids have been defined as substances having the following characteristics :-
- They are insoluble in water (hydrophobic) but soluble in non-polar solvents commonly termed fat solvents e:g ether , chloroform , benzene and acetone etc . This is because although certain lipids contain ionized groups (eg phosphate , serine and choline) , but the **bulk of any lipid is non-polar** .
- Their primary building blocks are fatty acids , glycerol , sphingosine and sterols .

Definition

- “Any of a class of organic compounds that are fatty acids or their derivatives and are insoluble in water but soluble in organic solvents”. .
- Essential structural component of living cells (along with proteins and carbohydrates), that helps shape, protect, and insulate the cell. It also provides an energy source..
- Lipids are a broad group of naturally-occurring molecules which includes fats and fats related substances, waxes, sterols, fat-soluble vitamins (such as vitamins A, D, E and K), natural oil , monoglycerides, diglycerides, phospholipids, and others.
- It contains hydrogen, carbon, and oxygen.

Classification of Lipids



Emulsification Test

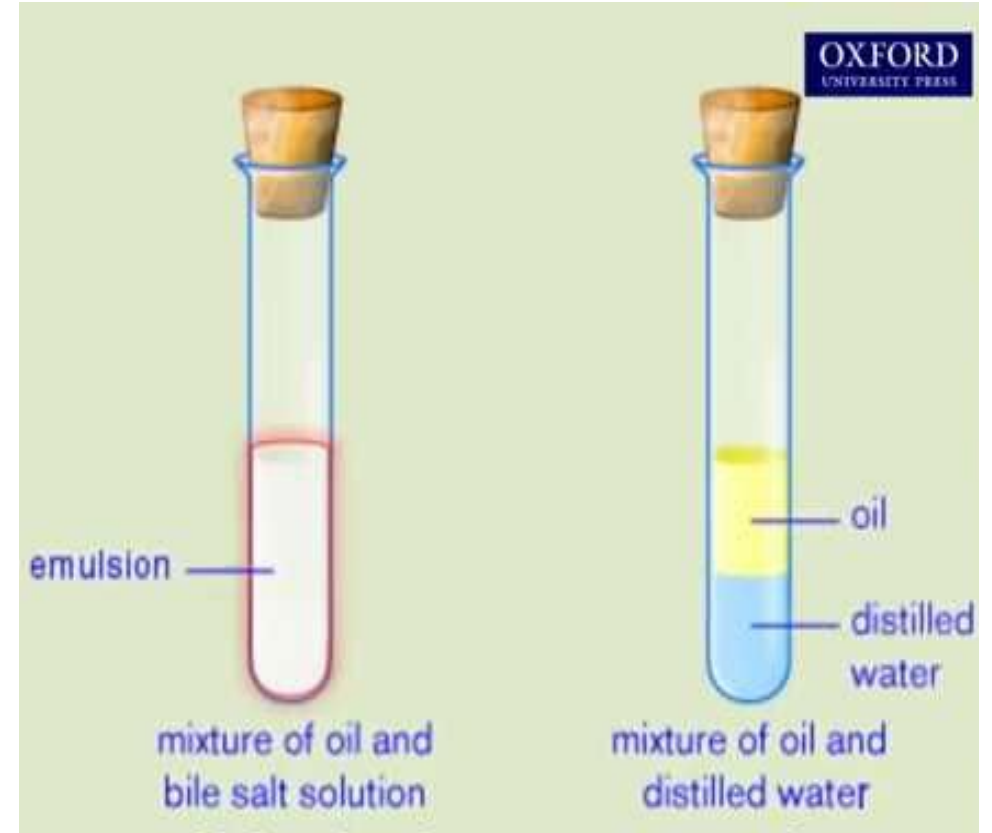
- ❑ **Objective:** To detect the presence of lipid in the given sample.
- ❑ **Principle:** The emulsification test is a commonly used method to determine the presence of lipids in any sample. Emulsification of fats means breaking up of fat globules into much smaller emulsion droplets and make it stable by emulsifying agent. Fat emulsification is important in the process of digestion in the intestine where different intestinal enzymes along with bile salts play their role. Emulsifiers are substances that reduce surface tension at interface of two normally immiscible phases allowing them to mix.

Requirements

- Given sample of oil
- Distilled water
- Emulsifying agent
- Test tubes and pipette

Procedure

- Take 2 ml of water in a test tube A, add 2-4 drops of given oil sample and shake well for about a minute. Note the appearance of solution.
- Take 2 ml of sample with emulsifying agent in test tube B, add 2-4 drops of given oil sample and shake well for about a minute.



DETECTION OF BILE SALT

- ❑ **Object:** To detect bile salts in the given sample by Hay's test.
- ❑ **Principle:** When sulfur powder is added to bile salts, they lowers the surface tension and allow the sulfur powder to sink.
- ❑ **Requirement:**
 - ❑ Sample
 - ❑ Sulfur powder
 - ❑ Clean & dry test tubes
 - ❑ Pipettes

□ **Procedure:**

- Take 2 test tubes and mark them
- In first tube take 5 ml distilled and sprinkle sulfur powder on surfaces.
- In second tube take 5ml sample and sprinkle sulfur powder on surfaces.

□ **Observation:**

- In first tube powder remains on surface of water.
- In second tube powder sink and settle at bottom.

Test for Bile salts

In the control, sulphur powder remains immiscible with the underlying liquid. In the positive test, the sulphur powder sinks to the bottom.

Interpretation: Bile salts and bile pigments are present in urine in obstructive jaundice.



Control for comparison



Positive test



THANK YOU